

UNIT 1

SYSTEM UNIT



Learning Objectives

- Define the system unit and describe its role in performing operations and housing key components.
- Explain the function of the motherboard and its role in connecting and enabling communication between computer parts.
- Describe the CPU (microprocessor) and its role as the computer's processing unit, explaining how it performs calculations.
- Identify and explain the five key components of the CPU: Arithmetic Logic Unit (ALU), Control Unit (CU), Clock, Registers, and Cache.



Computers are made up of both software and hardware. Software refers to the programs and applications that operate on a computer, while hardware includes the physical components that run these programs. Unlike software, hardware can be seen and touched. The key parts of computer hardware include the CPU (Central Processing Unit), memory, motherboard, graphics card, sound card, and the computer casing that holds everything together.



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The System Unit is the part of a computer that contains the primary devices. The system unit performs operations and produces results for complex calculations. It includes the motherboard, CPU, RAM, and other components, as well as the case in which these devices are enclosed.

1. Motherboard

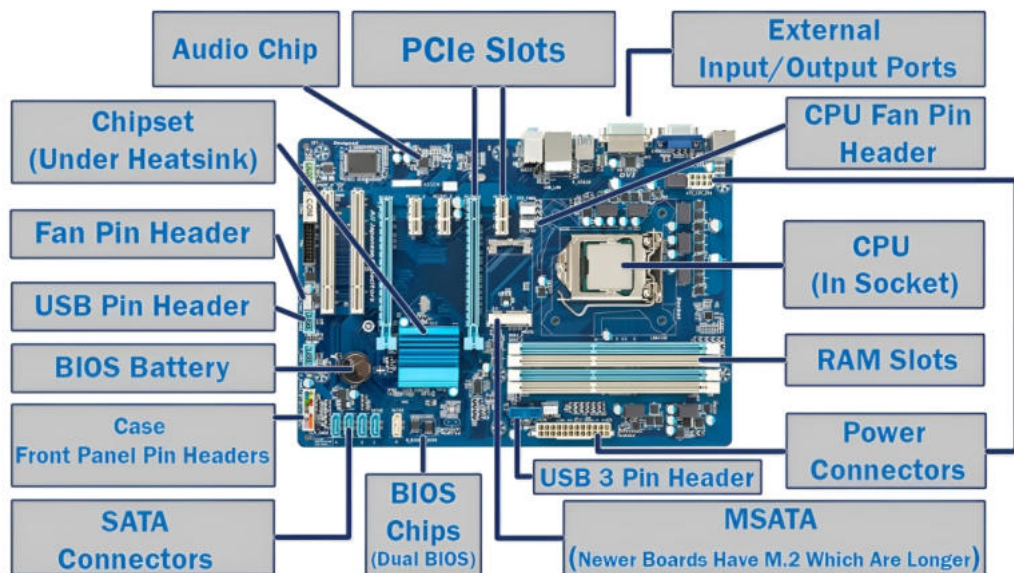
The motherboard is the main circuit board that connects all parts of a computer. It contains key components like the CPU, RAM, ROM, ports, and expansion slots. Similar to a nervous system, the motherboard enables communication between all parts of the computer.

Made of layered fiberglass, it contains copper circuits that carry electrical signals to different components. Devices such as input and output peripherals can be attached, removed, or upgraded through its ports as needed by the user.

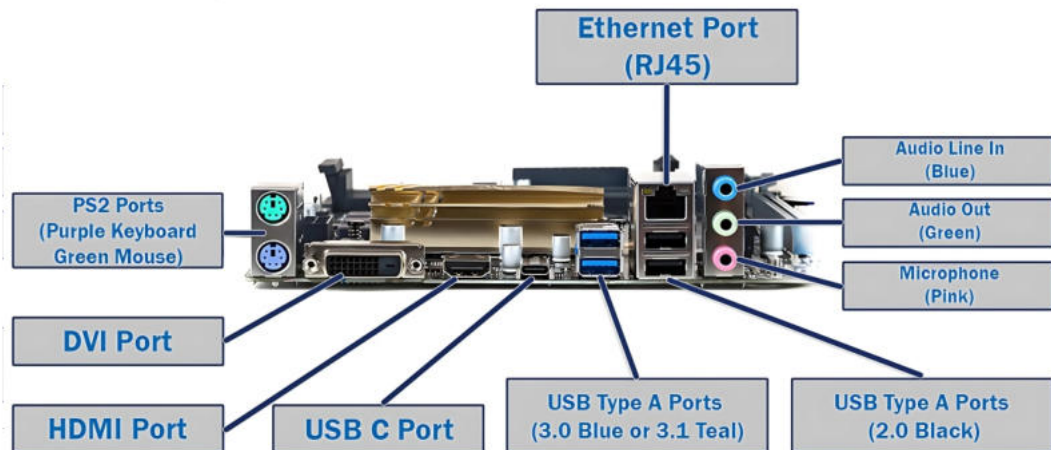


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Parts Of A Motherboard



External Input/Output Ports or Rear Connectors Of A Motherboard

2. Microprocessor

The CPU or microprocessor is the brain of the computer, made up of a silicon chip containing millions of transistors that process data. It performs all the calculations needed to operate the computer with high speed and accuracy.





The CPU (microprocessor) has five key components:

1. ARITHMETIC LOGIC UNIT (ALU):

Handles arithmetic operations (addition, subtraction, etc.) and logical comparisons (comparing, selecting, matching data).

2. CONTROL UNIT (CU):

Manages the flow of data and instructions between other computer units, coordinating operations like a traffic controller.

3. CLOCK:

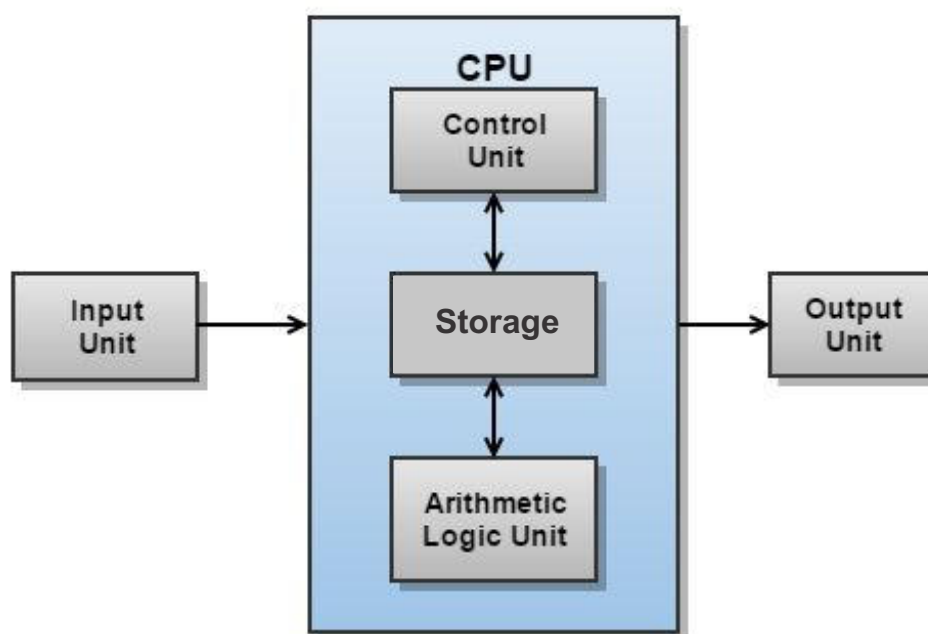
Generates pulses that dictate the timing of instruction execution, with speed measured in MHz or GHz.

4. REGISTERS:

Temporary storage for data being processed, available in sizes of 8, 16, 32, or 64 bits.

5. CACHE:

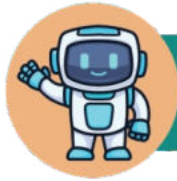
A fast, intermediate storage inside the microprocessor for temporarily holding processed information, available as internal (inside the CPU) or external (outside the CPU) cache.





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EXERCISE 1

Choose the correct answer.

1. What is the main function of the motherboard?

- A To connect and enable communication between all computer parts**
- B To store data**
- C To perform calculations**

2. The microprocessor is made up of:

- A Plastic**
- B Silicon**
- C Copper**

3. Which part of the CPU performs arithmetic and logic operations?

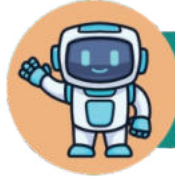
- A Control Unit**
- B Registers**
- C Arithmetic Logic Unit**

4. Which of the following is responsible for coordinating all parts of the computer?

- A CPU**
- B Control Unit**
- C ALU**

5. What is the speed of the CPU measured in?

- A Bits**
- B Bytes**
- C MHz or GHz**



EXERCISE 2



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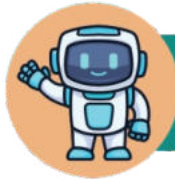
Fill in the blanks.

1. The _____ is the main circuit board that connects all parts of the computer.
2. The CPU, also known as the _____, performs all the calculations in the computer.
3. The _____ stores and processes data temporarily, while the computer is running.
4. A _____ enables communication between all parts of the computer, similar to a nervous system.
5. The speed of the CPU is measured in _____.



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EXERCISE 3

Mark "True" or "False".

TRUE

FALSE

The system unit contains all the primary devices needed to perform operations and calculations.

☐☐

Software refers to the physical components of a computer that can be seen and touched.

☐☐

The motherboard is the main circuit board that connects and enables communication between all parts of the computer.

☐☐

The Arithmetic Logic Unit (ALU) in the CPU is responsible for handling arithmetic operations and logical comparisons.

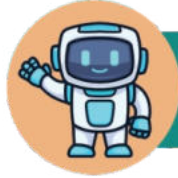
☐☐

The CPU has only one component that handles all its operations.

☐☐

Cache memory in the CPU is a temporary and fast storage used to hold processed information.

☐☐



EXERCISE 4



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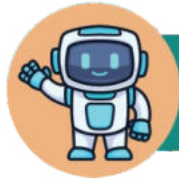
Answer the following questions.

- 1. What is the role of the microprocessor in a computer?**
- 2. Explain the function of CU.**
- 3. Explain the function of the motherboard.**
- 4. What is the purpose of the cache in the CPU?**
- 5. How does the Control Unit manage the operations inside the CPU?**



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[illegible]



EXERCISE 5



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Match the components of the system unit with their functions.

Column A

CPU

ALU

Motherboard

Registers

Control Unit

Column B

Temporary storage area

Handles arithmetic and logic operations

Main circuit board of the computer

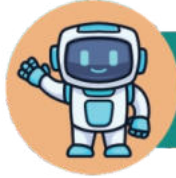
Brain of the computer

Controls the flow of data and instructions



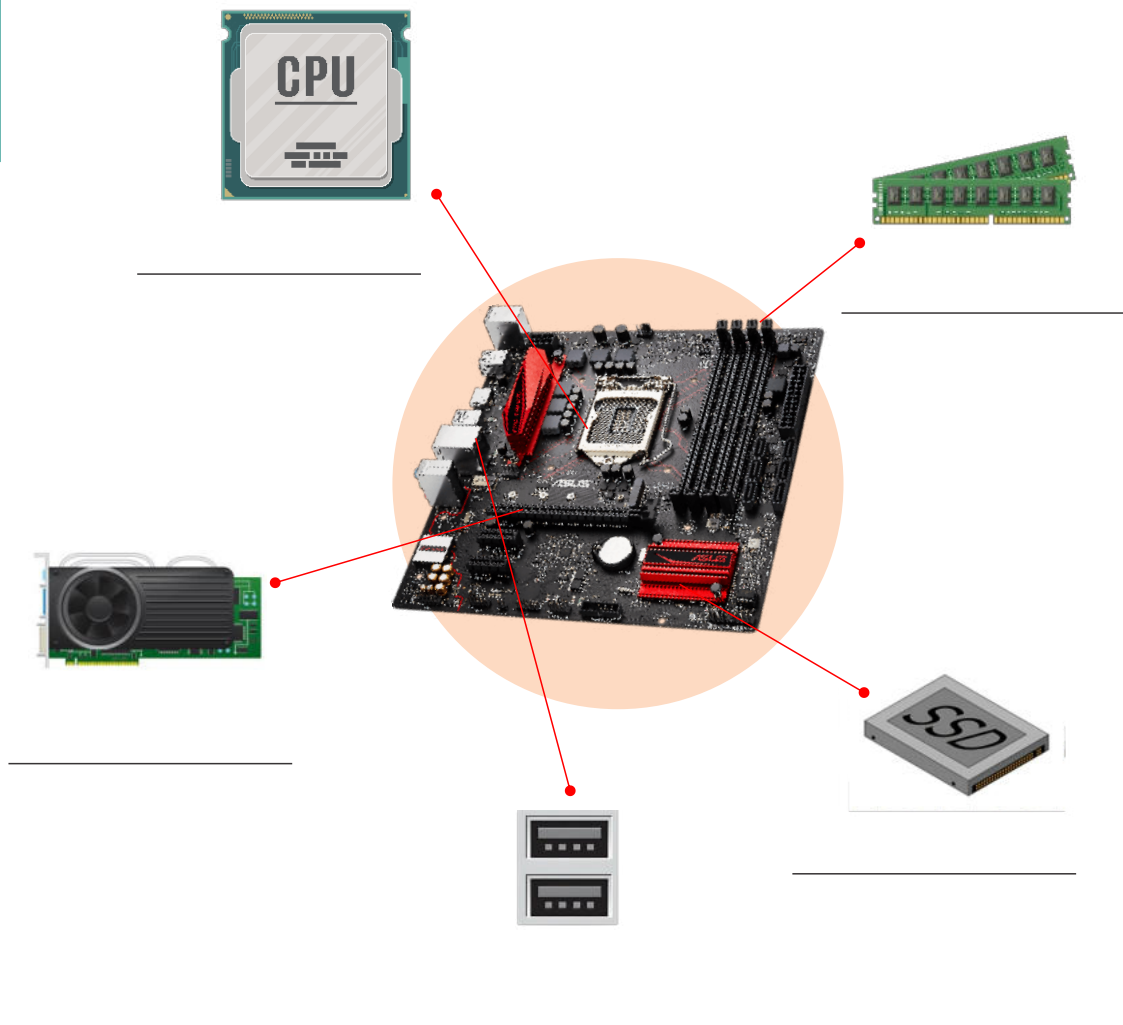
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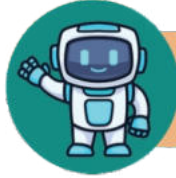
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EXERCISE 6

Label the diagram:



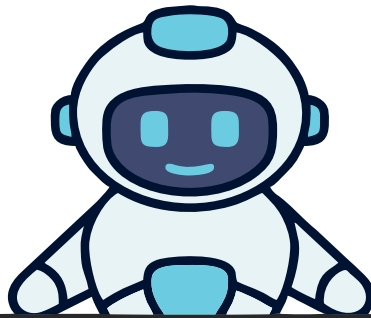


LAB ACTIVITY



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Using the available hardware, assemble a computer by correctly placing the following components: CPU, RAM, Motherboard, Power Supply, and Storage Device. After assembling, test the setup (real or virtual) to ensure it works correctly. Identify any issues if the computer doesn't function properly, and explain how you would fix them.

Follow up Questions:

1. If the computer doesn't start, what are the possible issues you would check first?
2. How would you identify whether the RAM is not functioning properly?
3. What happens if the power supply is not connected properly to the motherboard?
4. What part of the assembly process did you find the most challenging? How did you solve it?
5. If you had to upgrade one component in this computer, which one would it be and why?